

Section 3.9 Related Rates

$$\text{slope of tangent line} = \frac{dy}{dx} \begin{array}{l} \leftarrow \text{change of } y \\ \leftarrow \text{change of } x \end{array}$$

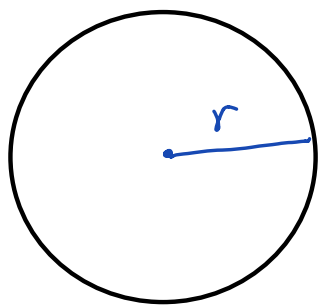
What about other derivatives?

If $A = \text{area}$, then

$$\frac{dA}{dt} \begin{array}{l} \leftarrow \text{change of Area} \\ \leftarrow \text{change of time} \end{array}$$

$$\frac{dA}{dr} \begin{array}{l} \leftarrow \text{change of Area} \\ \leftarrow \text{change of radius} \end{array}$$

Ex ① A fire starts at one point and spreads out in a circular pattern so that the radius grows by 2 feet per minute. How fast is the burn area increasing after 10 min.?



Given $\frac{dr}{dt} = 2$ and $t = 10$

Find $\frac{dA}{dt}$

$$A = \pi \cdot r^2$$

$$\frac{dA}{dt} = 2\pi \cdot r \cdot \frac{dr}{dt}$$

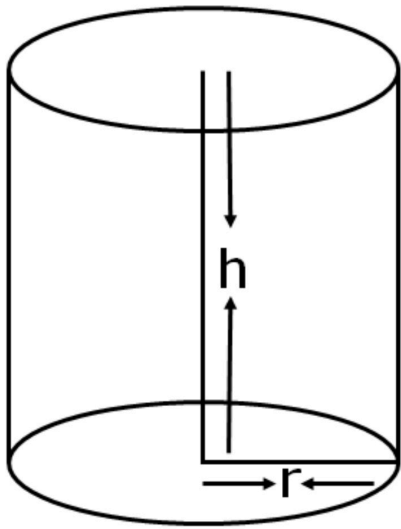
We need to know
 r when $t = 10 \text{ min.}$

$$r = 2 \frac{\text{feet}}{\text{min}} \cdot 10 \text{ min} = 20 \text{ feet}$$

$$\text{So, } \frac{dA}{dt} = 2\pi(20) \cdot 2$$

$$\frac{dA}{dt} = 251.327 \frac{\text{ft}^2}{\text{min}}$$

Ex ② A water tank is the shape of a right circular cylinder with a 3 foot radius. Water is draining out of the tank at 14 ft^3 per minute. How fast is the water level dropping?



Given $\frac{dV}{dt} = -14 \frac{\text{ft}^3}{\text{min}}$

And $r = 3 \text{ feet.}$

Find $\frac{dh}{dt}$

Volume $= \pi r^2 \cdot h$

$$V = \pi(3)^2 \cdot h$$

NOTE If a variable is NOT going to change, like $r=3$, you can substitute it into the formula before taking the derivative.

If a variable is going to change, like V , you must take the derivative first and then substitute the number.

$$V = 9\pi \cdot h$$

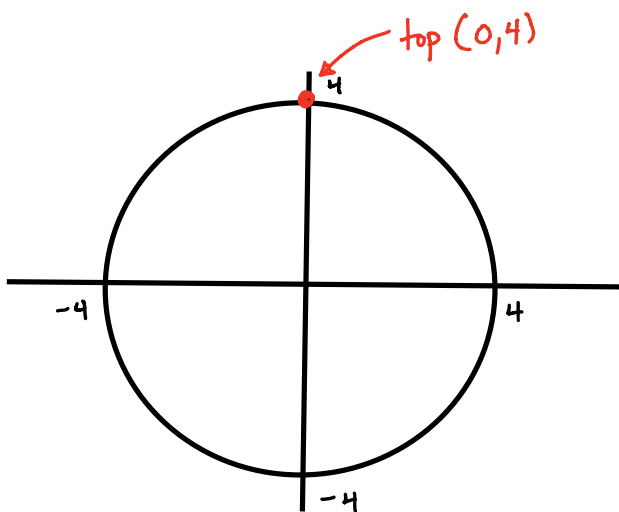
$$\frac{dV}{dt} = 9\pi \cdot \frac{dh}{dt}$$

$$-14 = 9\pi \cdot \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{-14}{9\pi} \text{ or } -0.495 \frac{\text{ft}}{\text{min}}$$

Ex ③ An object travels around a circle of radius 4 m. with a horizontal velocity of 12 cm per second. Find the vertical velocity when it is at the top of the path.

Note The units must be the same.
Change 12 cm to 0.12 m.



Circle
 $x^2 + y^2 = 16$

Given $\frac{dx}{dt} = 0.12$

Find $\frac{dy}{dt}$ At (0,4)

$$x^2 + y^2 = 16$$

$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 0$$

$$2y \cdot \frac{dy}{dt} = -2x \cdot \frac{dx}{dt}$$

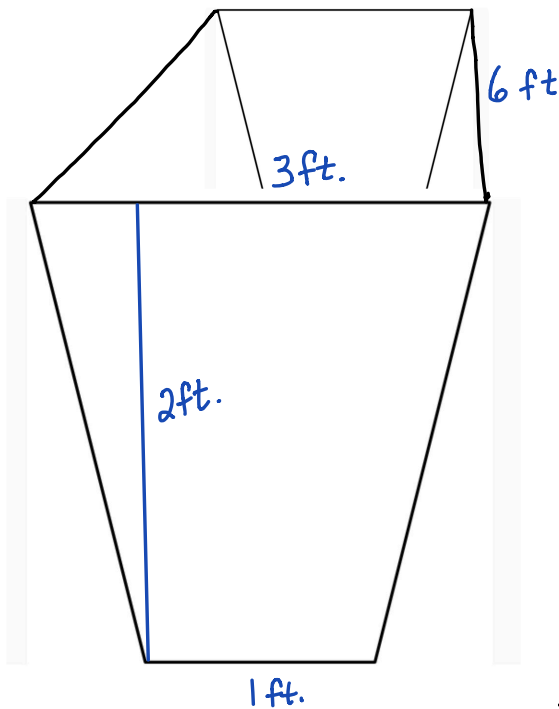
$$\frac{dy}{dt} = - \frac{x \cdot \frac{dx}{dt}}{y}$$

$$\frac{dy}{dt} = - \frac{0 \cdot (0.12)}{4} = 0 \frac{m}{sec}$$

So, at the top of the path
the object is not moving up or down.

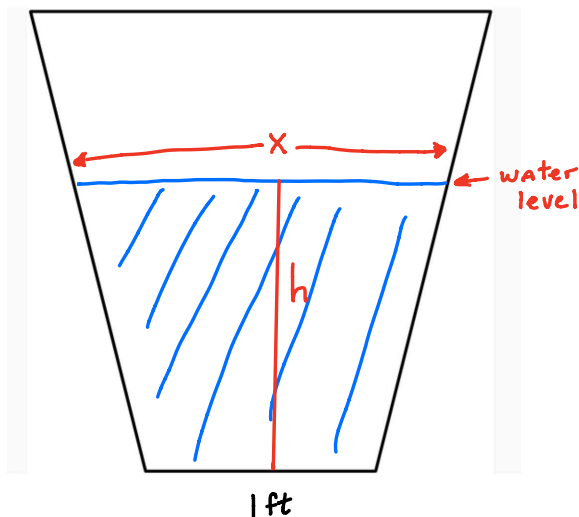
Ex ④ A water trough is 6 ft. long and 2 ft. tall.
Its ends are isosceles trapezoids
with base lengths 1 foot and 3 feet.
Water is pumped into the trough
at $20 \frac{ft^3}{min}$. How fast is the water

level rising After 5 mm?



This is the front view before any water has been Added.

As water rises, the 6ft and 1ft will NOT change, but the 3ft and 2ft need to be replaced with variables.



Here is the front face AS the water rises.

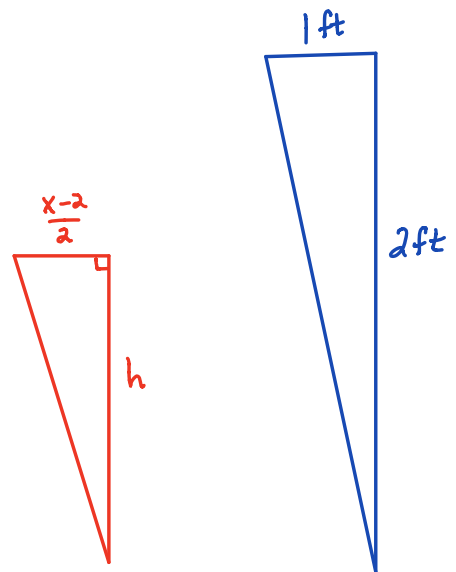
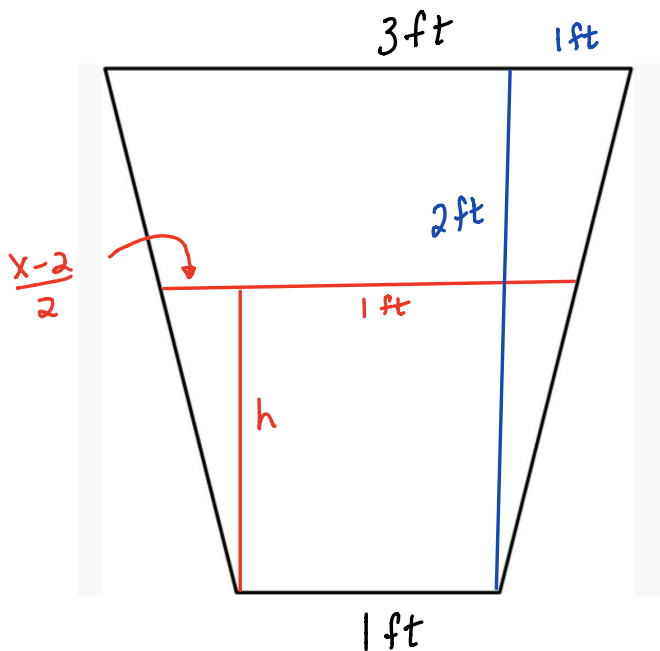
Volume of water
(Area of trapezoid) $\cdot$ 6ft

$$\text{Area of a trapezoid} = \frac{1}{2} (b_1 + b_2) \cdot h$$

$$\text{Area} = \frac{1}{2} (1+x) \cdot h$$

We are looking for $\frac{dh}{dt}$, but not $\frac{dx}{dt}$.

So, we need a way to replace x in this formula. Use similar triangles to do this.



$$\text{So, } \frac{\frac{x-2}{2}}{1} = \frac{h}{2}$$

Now solve for x .

$$\frac{x-2}{2} = \frac{h}{2}$$

$$x-2 = h$$

$$x = h+2$$

Substitute this into the Area formula.

$$\text{Area} = \frac{1}{2} (1+x) \cdot h$$

$$\text{Area} = \frac{1}{2} (1+h+2) \cdot h$$

$$A = \frac{1}{2} h^2 + \frac{3}{2} h$$

This means that the volume of water is

$$V = \left(\frac{1}{2} h^2 + \frac{3}{2} h \right) \cdot 6$$

$$V = 3h^2 + 9h$$

Now take the derivative.

$$\frac{dV}{dt} = 6h \cdot \frac{dh}{dt} + 9 \cdot \frac{dh}{dt}$$

We know $\frac{dV}{dt} = 20$ and $t = 5 \text{ min.}$,
but what does h equal?

$$V = 20 \frac{\text{ft}^3}{\text{min}} \cdot 5 \text{ min} = 100 \text{ ft}^3$$

$$V = 3h^2 + 9h$$

$$100 = 3h^2 + 9h$$

Use Quadratic Formula to get $h = 4.465 \text{ ft}$

Finally, substitute $\frac{dV}{dt} = 20$ and $h = 4.465$

into
$$\frac{dV}{dt} = 6h \cdot \frac{dh}{dt} + 9 \cdot \frac{dh}{dt}$$

$$20 = 6(4.465) \frac{dh}{dt} + 9 \cdot \frac{dh}{dt}$$

$$20 = 35.79 \cdot \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{20}{35.79} = 0.559 \frac{ft}{min}$$